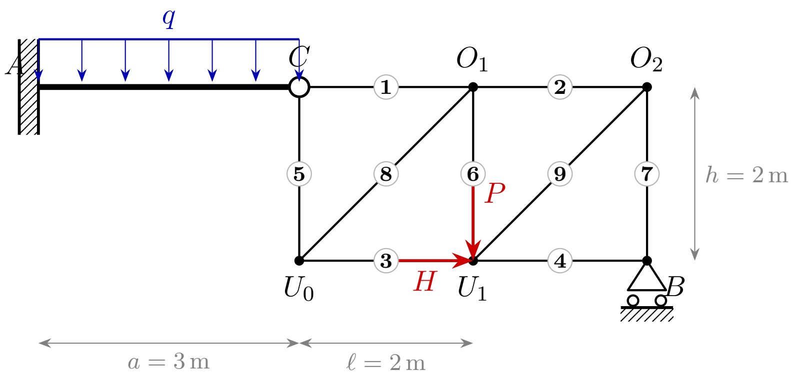
University of Natural Resources and Life Sciences, Vienna (BOKU)

VU Mechanik – LAWI 100515 (PI)

Exam **Group A** Date: 26.06.2026

Given system

The system consists of two rigid bodies connected by a hinge at C: a bending beam A–C (fixed support at A) and a truss (roller support at B).



Given quantities: $a = 3\,\mathrm{m}$ $\ell = 2\,\mathrm{m}$ $h = 2\,\mathrm{m}$ $q = 6\,\mathrm{kN/m}$ $P = 20\,\mathrm{kN}$ $H = 5\,\mathrm{kN}$

Tasks

1. Determine the degree of static determinacy of the system.
2. Compute all support reactions and the hinge force at C.
3. Determine all member forces of the truss (state tension/compression).
4. Draw the section-force diagrams $N(x)$, $V(x)$ and $M(x)$ for the bending beam A–C.